

ALGEBRA PROPERTIES

ARITHMETIC PROPERTIES

ASSOCIATIVE	$a(bc) = (ab)c$
COMMUTATIVE	$a + b = b + a$ and $ab = ba$
DISTRIBUTIVE	$a(b + c) = ab + ac$

ARITHMETIC OPERATIONS EXAMPLES

$$\begin{aligned} ab + ac &= a(b + c) & \frac{a}{b} - \frac{c}{d} &= \frac{ad - bc}{bd} \\ a\left(\frac{b}{c}\right) &= \frac{ab}{c} & \frac{a - b}{c - d} &= \frac{b - a}{d - c} \\ \left(\frac{a}{b}\right) &= \frac{a}{bc} & \frac{a + b}{c} &= \frac{a}{c} + \frac{b}{c} \\ \frac{a}{\left(\frac{b}{c}\right)} &= \frac{ac}{b} & \frac{ab + ac}{a} &= b + c, a \neq 0 \\ \frac{a}{b} + \frac{c}{d} &= \frac{ad + bc}{bd} & \left(\frac{a}{b}\right) &= \frac{ad}{bc} \\ & & \left(\frac{c}{d}\right) &= \frac{bc}{ad} \end{aligned}$$

QUADRATIC EQUATION

For the equation $ax^2 + bx + c = 0$

$$x = \frac{-b \pm \sqrt{b^2 - 4ac}}{2a}$$

RADICAL PROPERTIES

$a, b \geq 0$ for even n

$$\begin{aligned} \sqrt[n]{a} &= a^{\frac{1}{n}} \\ \sqrt[m]{\sqrt[n]{a}} &= \sqrt[mn]{a} \\ \sqrt[n]{ab} &= \sqrt[n]{a}\sqrt[n]{b} \\ \sqrt[n]{\frac{a}{b}} &= \frac{\sqrt[n]{a}}{\sqrt[n]{b}} \\ \sqrt[n]{a^n} &= a, \text{ if } n \text{ is odd} \\ \sqrt[n]{a^n} &= |a|, \text{ if } n \text{ is even} \end{aligned}$$

LOGARITHM PROPERTIES

if $y = \log_b x$ then $b^y = x$

$$\begin{aligned} \log_b b &= 1 \text{ and } \log_b 1 = 0 \\ \log_b b^x &= x \\ b^{\log_b x} &= x \\ \log_a x &= \frac{\log_b x}{\log_b a} \\ \log_b(x^r) &= r \log_b x \\ \log_b(xy) &= \log_b x + \log_b y \\ \log_b\left(\frac{x}{y}\right) &= \log_b x - \log_b y \end{aligned}$$

EXPONENT PROPERTIES

$$\begin{aligned} a^n a^m &= a^{n+m} \\ (a^n)^m &= a^{nm} \\ (ab)^n &= a^n b^n \\ a^{-n} &= \frac{1}{a^n} \\ \left(\frac{a}{b}\right)^{-n} &= \left(\frac{b}{a}\right)^n = \frac{b^n}{a^n} \\ \frac{a^n}{a^m} &= a^{n-m} = \frac{1}{a^{m-n}} \\ a^0 &= 1, a \neq 0 \\ \left(\frac{a}{b}\right)^n &= \frac{a^n}{b^n} \\ \frac{1}{a^{-n}} &= a^n \\ a^{\frac{n}{m}} &= \left(a^{\frac{1}{m}}\right)^n = (a^n)^{\frac{1}{m}} \end{aligned}$$

PROPERTIES OF INEQUALITIES

If $a < b$ then $a + c < b + c$ and $a - c < b - c$
If $a < b$ and $c > 0$ then $ac < bc$ and $a/c < b/c$
If $a < b$ and $c < 0$ then $ac > bc$ and $a/c > b/c$

PROPERTIES OF COMPLEX NUMBERS

$$\begin{aligned} i &= \sqrt{-1} \\ i^2 &= -1 \\ \sqrt{-a} &= i\sqrt{a}, \quad a \geq 0 \\ (a + bi) + (c + di) &= a + c + (b + d)i \\ (a + bi) - (c + di) &= a - c + (b - d)i \\ (a + bi)(c + di) &= ac - bd + (ad + bc)i \\ (a + bi)(a - bi) &= a^2 + b^2 \\ |a + bi| &= \sqrt{a^2 + b^2} \\ \overline{(a + bi)} &= a - bi \\ \overline{(a + bi)}(a + bi) &= |a + bi|^2 \\ \frac{1}{(a + bi)} &= \frac{(a - bi)}{(a + bi)(a - bi)} = \frac{a - bi}{a^2 + b^2} \end{aligned}$$

COMMON FACTORING EXAMPLES

$$\begin{aligned} x^2 - a^2 &= (x + a)(x - a) \\ x^2 + 2ax + a^2 &= (x + a)^2 \\ x^2 - 2ax + a^2 &= (x - a)^2 \\ x^2 + (a + b)x + ab &= (x + a)(x + b) \\ x^3 + 3ax^2 + 3a^2x + a^3 &= (x + a)^3 \\ x^3 + a^3 &= (x + a)(x^2 - ax + a^2) \\ x^3 - a^3 &= (x - a)(x^2 + ax + a^2) \\ x^{2n} - a^{2n} &= (x^n - a^n)(x^n + a^n) \end{aligned}$$

ABSOLUTE VALUE

$$\begin{aligned} |a| &= \begin{cases} a, & \text{if } a \geq 0 \\ -a, & \text{if } a < 0 \end{cases} \\ |a| &= |-a| \\ |a| &\geq 0 \\ |ab| &= |a||b| \\ \left|\frac{a}{b}\right| &= \frac{|a|}{|b|} \\ |a + b| &\leq |a| + |b| \end{aligned}$$

COMPLETING THE SQUARE

$$ax^2 + bx + c = a(\dots)^2 + \text{constant}$$

1. Divide by the coefficient a .
2. Move the constant to the other side.
3. Take half of the coefficient b/a , square it and add it to both sides.
4. Factor the left side of the equation.
5. Use the square root property.
6. Solve for x .